

[Kat]herine Titterton

San Francisco, CA · macrofluidic.me · github.com/kat-titter · X @KatTitterton

TL;DR Scientist-engineer at the seam of imaging, machine learning, and lab automation in biotech. 8+ years turning bench work into machine-learning fuel: building and scaling high-content assays, then the image analysis that unlocks phenotypic discovery at scale. Part process engineer, part artist-technologist who finds the shortest path and walks it.



[MACROFLUIDIC.ME/RESUME](https://macrofluidic.me/resume)

EXPERIENCE

• Peer-reviewed publications — full list on page 2

Founding Member of the Technical Staff, Axiom Bio · axi.om

2023–Present · San Francisco, CA

Founding technical hire building new approach methodologies (NAMs) for drug safety. Led feature engineering and algorithm development; oversaw data generation for two of the world's largest Cell Painting datasets (100,000+ compounds). Invented two bioactivity-detection methods ~10× more sensitive than conventional toxicity assays. Honorary Artist in Residence.

• *Nat Commun* 2026 · *Cell Syst* 2026

Associate Scientist III, Insitro · Arrayed Screening, Neuroscience

2022–2023 · South San Francisco, CA

Adapted high-content phenotypic assays for high-throughput small-molecule screening (TSC, ALS) on near-fully-automated platforms (BioTek, High-Res). Built a digital phase-contrast timecourse imaging modality and Python analysis suite for label-free phenotypic discovery.

Technical Associate II, Massachusetts Institute of Technology · Synthetic Neurobiology Group

2021–2022 · Cambridge, MA

Optimized multiplexed expansion revealing (mExR) for 7+ rounds of IHC on 16× expanded tissue, imaging 30+ proteins at nanoscale. Built image-analysis pipelines in ImageJ/Fiji and MATLAB.

• *Nat Commun* 2024

Associate Scientist II, AbbVie · Foundational Neuroscience Center, Early R&D

2018–2021 · Cambridge, MA

Built primary murine and hiPSC neuronal tau-aggregation high-content screening assays (Revvity Opera, Harmony). Modeled tau propagation in Alzheimer's disease with microfluidic devices; authored R dose-response pipelines.

• *Mol Neurodegener Adv* 2026 · *iScience* 2026 · *J Biol Chem* 2025 · *iScience* 2022

Research Technician I, Massachusetts General Hospital

2017–2018 · Boston, MA

Used voltage-clamp electrophysiology in GABAR-expressing *Xenopus* oocytes to elucidate mechanisms of general anesthetics.

• *Eur J Med Chem* 2019

PROJECTS

Bare Metal

2025

First IRL art exhibit, Human Tech Week, SF. Post-hoc analysis on macrofluidic.me/blog.

Vivarium 2.0

2024

Rebuilt habitat for Axiom Bio's poison dart frogs: automated misting and a motion-triggered camera streaming live frog video.

OpenTransformers (NYC)

2023

LLM code-generation companion for the OT-2 lab robot and its Python SDK. 2nd place, BioHack NYC 2023.

EDUCATION

Harvard University · Extension School, graduate coursework

2017–2020

Completed while employed full-time: Social Medicine; Intro to Programming (Python); Image Processing & Computer Vision (Python).

B.A., Colgate University · *cum laude*

2013–2017

Major: Cellular Neuroscience. Minor: Mathematical Systems Biology. Co-captain, Division I Track & Field; Dorothy-Waltham Award for Athletic Leadership.

SKILLS & TOOLS

COMPUTATIONAL

LLM codegen (Claude Code, Cursor) · image embeddings (clustering, feature derivation, normalization) · Python (NumPy, SciPy, scikit-learn, pandas, matplotlib, seaborn; image processing & segmentation) · R (tidyverse, ggplot2, drc, platetools) · dose-response fitting (Prism, R, Python) · multivariate statistics · design of experiments · ImageJ/Fiji macros · Benchling ELN.

WET LAB

Primary murine & hiPSC neuron culture · microfluidic cell-culture models · ICC / IHC · confocal & high-content imaging (Opera Phenix / Operetta, Harmony) · automated liquid handling & scheduling (OpenTrons, Integra, BioTek, Labcyte Echo, High-Res Biosolutions, Green Button Go) · assay development & HT/HC screening · stereotaxic surgery / microdissection · voltage-clamp electrophysiology.

CERTIFICATIONS

FII Level 1 Freediving (Oct 2019) · Molchanov L2 Freediving (Oct 2025) · SII Shark Ecology Specialty (Apr 2020)

PEER-REVIEWED PUBLICATIONS

8 papers · newest first

1. Yanamandra K, Romanul N, Patel H, Sanders H, Chang R, Darcy A, Brown N, Kane C, Wu HY, Jagtap K, Chancellor S, Morozova O, Xi M, Hogan C, **Titterton K**, Wu JW, Manos J, Karran E, Dellovade T, Langlois X. Conformation-specific anti-pathological tau antibody raised against AD brain-derived seed-competent tau reduces tau pathology in transgenic mice. *Mol Neurodegener Adv*. 2026;2(1):43. doi:10.1186/s44477-026-00049-6
2. Sebastian RM, Kupp R, Nanavati D, Parfentev I, Krzystek TJ, Romanul N, Yanamandra K, Preiss CN, Manos JD, Wu HY, Volkova A, Crescenti S, **Titterton K**, Welker AM, Townsend M, Karran E, Langlois X, Wu JW. SynCAR platform for capturing neuronal synaptic proteopathic seeds. *iScience*. 2026;29(7):116362. doi:10.1016/j.isci.2026.116362 PMID 42325569; PMCID PMC13276770.
3. Ewald JD, **Titterton KL**, Bäuerle A, Beatson A, Boiko DA, Cabrera AA, Cheah J, Cimini BA, Gorissen BL, Harrill J, Jones TR, Karczewski KJ, Crute CE, Rouquie D, Seal S, Weisbart E, White B, Carpenter AE, Singh S. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Syst*. 2026;17(5):101566. doi:10.1016/j.cels.2026.101566 PMID 41903531.
4. Seal S, Dee W, Shah A, Cerisier N, Zhang A, Miglietta E, **Titterton K**, Cabrera AA, Boiko D, Beatson A, Slabaugh G, Taboureaux O, Carreras Puigvert J, Singh S, Spijth O, Bender A, Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*. 2026;17(1). doi:10.1038/s41467-026-68725-5 PMID 41651839; PMCID PMC12988037.
5. Wu JW, **Titterton K**, Sebastian R, Romanul N, Yanamandra K, Mastis B, Chang R, Khasnavis S, Kwon T, Chai Y, Manos JD, Preiss CN, Hu M, Welker AM, Liao F, Dellovade T, Karran E, Langlois X. Development of a robust, physiologically relevant neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem*. 2025;301(12):110853. doi:10.1016/j.jbc.2025.110853 PMID 41135680; PMCID PMC12670552.
6. Kang J, Schroeder ME, Lee Y, Kapoor C, Yu E, Tarr TB, **Titterton K**, Zeng M, Park D, Niederst E, Wei D, Feng G, Boyden ES. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nat Commun*. 2024;15(1):9722. doi:10.1038/s41467-024-53729-w PMID 39521775; PMCID PMC11550395.
7. Manos JD, Preiss CN, Venkat N, Tamm J, Reinhardt P, Kwon T, Wu J, Winter AD, Jahn TR, Yanamandra K, **Titterton K**, Karran E, Langlois X. Uncovering specificity of endogenous TAU aggregation in a human iPSC-neuron TAU seeding model. *iScience*. 2022;25(1):103658. doi:10.1016/j.isci.2021.103658 PMID 35072001; PMCID PMC8761709.
8. Wu B, Jayakar SS, Zhou X, **Titterton K**, Chiara DC, Szabo AL, Savechenkov PY, Kent DE, Cohen JB, Forman SA, Miller KW, Bruzik KS. Inhibitable photolabeling by neurosteroid diazirine analog in the $\beta 3$ -subunit of human heteropentameric type A GABA receptors. *Eur J Med Chem*. 2019;162:810-824. doi:10.1016/j.ejmech.2018.11.019 PMID 30544077; PMCID PMC6912599.

CONFERENCE PRESENTATIONS & POSTERS

1. **Titterton K**, Boiko D, Cabrera AA, Bäuerle A, Tyagi A, Saneinejad S, Beatson A, White B. "Accurate and interpretable in silico clinical risk assessment for drug-induced liver injury (DILI) from molecular structure." *Society of Toxicology (SOT) Annual Meeting*, 2025.
2. **Titterton K**, et al. "Development of a tau propagation neuronal microfluidic cell culture model for assessing inhibitors of tau propagation." *Society for Neuroscience (SfN) Global Connectome Virtual Meeting*, January 2021.
3. **Titterton K**, et al. "High content analysis (HCA) of tau aggregation in primary neurons: screening anti-tau antibodies and modeling tauopathy progression in vitro." *SBI2 High Content Meeting*, September 2019, Harvard Medical School, Boston, MA.

Earlier research presentations

- "Investigating RNAi as a potential therapeutic for Huntington's Disease in a transgenic mHTT145Q zebra finch (*Taeniopygia guttata*) model." Colgate University Neuroscience Thesis Poster Presentations, 2017.
- "Atopy and helminth infection in Ethiopian schoolchildren." Colgate Summer Research Poster Symposium, 2016.
- "PDE2 regulation of β -arrestin signaling." Systems Biology Center New York (SBCNY) Summer Trainee Poster Symposium, Icahn School of Medicine at Mount Sinai, 2015.